



Maintenance Schedule

Diesel Engine

V2000S56

V2000S96

V2000C66

V2000C66R

Application Groups

4B, 4D, 5B

MS50090/06E

Engine model	kW/cyl.	Application group
12V2000S56	65 kW/cyl.	4B, Continuous operation, variable
16V2000S56	60 kW/cyl.	4B, Continuous operation, variable
12V2000S96	71 kW/cyl.	4D, Frac operation (continuous operation at low load)
16V2000S96	72 kW/cyl.	4D, Frac operation (continuous operation at low load)
12V2000C66	65 kW/cyl.	5B, Continuous operation, variable
12V2000C66R	65 kW/cyl.	5B, Continuous operation, variable
16V2000C66	60 kW/cyl.	5B, Continuous operation, variable

Table 1: Applicability

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1 Maintenance Schedule

1.1 Preface

This manual provides instructions for your product from the manufacturer Rolls-Royce Solutions.

Maintenance concept

The maintenance system for Rolls-Royce Solutions products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals as well as the required scopes of maintenance tasks are based on operational experience and therefore to be considered as recommendations. Additional maintenance work and/or changes to the maintenance intervals may be required in the case of difficult operational and ambient conditions.

Adherence to the specified maintenance intervals is essential to maintain product safety.

The intervals according to which the maintenance tasks have to be carried out are specified as operating hours and time limits. Whichever value is reached first shall apply.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by the Service Partner of Rolls-Royce Solutions. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product.

QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule).

QL3: Maintenance work which requires partial disassembly of the product.

QL4: Maintenance work which requires complete disassembly of the product.

Additional notes on maintenance and preservation:

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

For the change intervals of fluids and lubricants, refer to the corresponding Fluids and Lubricants Specifications.

If the product is to remain out of service for a longer time period, Rolls-Royce Solutions recommends to carry out a monthly test run until engine operating temperature is reached. If the engine is scheduled to remain out of service for > 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications

The latest version of the specifications is available at <http://www.mtu-solutions.com>.

1.2 Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

Interval (HRS)/limit	Limit in %	Limit max.
Operating hours	± 10%	± 500 hrs
Years	± 10%	± 30 days
Example 1	Task every 1000 hrs	Between 900 hrs and 1100 hrs (10% of 1000 hrs)
Example 2	Task every 30000 hrs	Between 29500 hrs and 30500 hrs (10% of 30000 hrs, but max. 500 hrs)

1.3 Allocation matrix for application groups 4B, 4D, 5B

Note:

TBO identifiable via cylinder power (shown on engine nameplate or refer to the engine data in the Business Portal, power per cylinder) and, where appropriate, in the contractually agreed load profile.

Allocation matrix for application group 4B

Load profile		Load factor %	12/16V2000-06 60 kW/cyl. 65 kW/cyl.
Load %	Time %		
110	1	75.1	10,000 h
100	29		
75	40		
50	30		
100	10	58.5	10,000 h
75	40		
50	30		
25	10		
10	10	45.75	12,000 h
90	5		
50	65		
35	25		
0	5		

Load profile		Load factor %	12/16V2000-06 60 kW/cyl. 65 kW/cyl.
Load %	Time %		
100	5	45	12,000 h
80	15		
55	40		
20	20		
10	20	44.25	12,000 h
100	5		
65	35		
40	25		
30	15	24.2	16,000 h
10	20		
100	2		
80	2		
50	20		
20	30		
10	46		

Allocation matrix for application group 4D

Load profile		Load factor %	12/16V2000-06	
Load %	Time %		60 kW/cyl. 65 kW/cyl.	71 kW/cyl. 72 kW/cyl.
100	5	45	14,000 h	12,000 h
80	15			
55	40			
20	20			
10	20			
100	0.4	43.15	16,000 h	14,000 h
90	2			
80	4.7			
70	7.5			
60	9.9			
50	19.6			
40	21.7			
30	17.8			
20	5.6			
10	10.6			

Load profile		Load factor %	12/16V2000-06	
Load %	Time %		60 kW/cyl. 65 kW/cyl.	71 kW/cyl. 72 kW/cyl.
100	2	24.2	16,000 h	15,000 h
80	2			
50	20			
20	30			
10	46			

Allocation matrix for application group 5B

Load profile		Load factor %	12/16V2000-06	
Load %	Time %		60 kW/cyl.	65 kW/cyl.
100	46	59.6	12,000 h	10,000 h
80	14			
20	12			
0	28			
100	40	54.4	14,000 h	12,000 h
80	15			
20	12			
0	33			

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Load profile		Load factor %	12/16V2000-06	
Load %	Time %		60 kW/cyl.	65 kW/cyl.
100	35	50.2	16,000 h	14,000 h
80	16			
20	12			
0	38			
100	30	45.2	16,000 h	14,000 h
80	16			
20	12			
0	42			
100	26	40.4	16,000 h	16,000 h
80	15			
20	12			
0	47			
100	21	35	18,000 h	18,000 h
80	14			
20	14			
0	51			
100	17	30.4	18,000 h	18,000 h
80	13			
20	15			
0	55			

Load profile		Load factor %	12/16V2000-06	
Load %	Time %		60 kW/cyl.	65 kW/cyl.
100	12	24.8	18,000 h	18,000 h
80	12			
20	16			
0	60			

1.4 Maintenance schedule matrix 18,000h

0-10,500 Operating hours

Item	Limit	Operating hours [h]																
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000
Fuel servicing plant	1 a																	
Fuel servicing plant	2 a																	
Engine oil filter	2 a																	
Engine operation	-	X																
Centrifugal oil filter	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Fuel filter	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Valve gear	2 a					X				X				X				X
Air filters	3 a							X					X					X
Belt drive	2 a										X							X
Crankcase breathers	6 a										X							X
NOx sensor	6 a										X							X
Oxygen sensor	6 a										X							X
Fuel injectors	9 a																	X
Pressure pipe neck (bayonet)	9 a																	X
Engine governor	9 a																	X
Exhaust gas recirculation	18 a																	X
Cylinder heads	9 a																	X
Turbochargers	9 a																	X
Exhaust gas recirculation	18 a																	X
Component maintenance	9 a																	X
HP fuel pumps	9 a																	
Extended component maintenance	18 a																	
w = weeks m = months a = years																		

11,000-18,000 Operating hours

Item	Limit	Operating hours [h]																
		11,000	11,500	12,000	12,500	13,000	13,500	14,000	14,500	15,000	15,500	16,000	16,500	17,000	17,500	18,000		
Fuel servicing plant	1 a																	
Fuel servicing plant	2 a																	
Engine oil filter	2 a																	
Engine operation	-																	
Centrifugal oil filter	2 a	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X		
Fuel filter	2 a	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X		
Valve gear	2 a			X				X				X				X		
Air filters	3 a			X						X						X		
Belt drive	2 a						X									X		
Crankcase breathers	6 a						X									X		
NOx sensor	6 a						X									X		
Oxygen sensor	6 a						X									X		
Fuel injectors	9 a															X		
w = weeks m = months a = years																		

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Item	Limit	Operating hours [h]															
		11,000	11,500	12,000	12,500	13,000	13,500	14,000	14,500	15,000	15,500	16,000	16,500	17,000	17,500	18,000	
Pressure pipe neck (bayonet)	9 a															X	
Engine governor	9 a															X	
Exhaust gas recirculation	18 a															X	
Cylinder heads	9 a															X	
Turbochargers	9 a															X	
Exhaust gas recirculation	18 a															X	
Component maintenance	9 a															X	
HP fuel pumps	9 a															X	
Extended component maintenance	18 a															X	
w = weeks m = months a = years																	

1.5 Maintenance tasks 18,000h

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
QL1	-	1 a	Fuel servicing plant	Check alarm function of differential pressure gauge.	X	W1245
				Check pump capacity.	X	W1246
QL1	-	2 a	Fuel servicing plant	Check function of bar electrode.	X	W1244
QL1	-	2 a	Engine oil filter	Fit new engine oil filters each time the engine oil is changed or, at the latest, on expiry of the time limit (given in years).		W1008
QL1	Daily	-	Engine operation	Check engine oil level.		W0500
				Carry out visual inspection of engine for general condition and leaks.		W0501
				Inspect intercooler drain system.	X	W0502
				Inspect service indicator of air filter.		W0503
				Check relief bores of coolant pump(s).		W0505
				Check for abnormal running noises, exhaust gas color and vibration.		W0506
				Drain off water and contamination from fuel prefilter.	X	W0507
				Check differential pressure gauge of fuel prefilter.	X	W0508
QL1	500	2 a	Centrifugal oil filter	Check thickness of oil residue layer. Clean. Fit new sleeve, at the latest, each time the engine oil is changed.	X	W1009
QL1	500	2 a	Fuel filter	Fit new fuel filter or new fuel filter insert.		W1001
QL1	2000	2 a	Valve gear	Check valve clearance, adjust if required. ATTENTION! First adjustment after 1,000 hours.		W1207
QL1	3000	3 a	Air filters	Fit new air filters.		W1005
QL1	4500	2 a	Belt drive	Check belt condition and tension Fit new belt(s) if necessary.		W1003
QL1	4500	6 a	Crankcase breathers	Fit new fine separator for oil mist.		W1547
QL1	4500	6 a	NOx sensor	Fit new sensor.		W1525
QL1	4500	6 a	Oxygen sensor	Fit new sensor.		W1526
QL1	9000	9 a	Fuel injectors	Replace fuel injectors.		W1006
QL1	9000	9 a	Pressure pipe neck (bayonet)	Fit new pressure pipe neck (bayonet) in cylinder head.		W1178
QL1	9000	9 a	Engine governor	Reset parameters of drift correction (CDC) and enter injector coding (IIG).		W1636
QL1	9000	18 a	Exhaust gas recirculation	Check operation of flaps and actuators.		W1523
QL3	9000	9 a	Cylinder heads	Overhaul cylinder heads. Visually inspect piston crowns and wear pattern on cylinder liner running surfaces.		W1063
QL3	9000	9 a	Turbochargers	Overhaul turbocharger (MTU-ZR-turbocharger) If Garrett turbocharger, fit new.		W1388
QL3	9000	18 a	Exhaust gas recirculation	Endoscopy EGR cooler, clean as needed.		W1524
QL3	9000	9 a	Component maintenance	Before starting maintenance work, drain the coolant and flush the cooling systems.		W2000
				Clean air ducting.		W2002
				Clean intercooler and inspect for leakage.		W2003
				Check engine coolant thermostat and fit new thermal actuator.		W2006
				Inspect centrifugal oil filter for wear.	X	W2009
				Clean engine oil cooler and inspect it for leaks.		W2018
				Check engine alignment.		W2036
				Inspect rocker arms, valve bridges, pushrods and ball joints for wear.		W2074
				Fit new seals/sealing materials for all disassembled components.		W2062
				Overhaul engine coolant pump.		W2110
				Overhaul charge-air coolant pump.		W2070
				Check actuating motors of EGR flaps, replace if necessary.		W2157
QL3	18000	9 a	HP fuel pumps	Fit new HP fuel pumps.		W1058

w = weeks
m = months
a = years

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